

Exp. No: 2

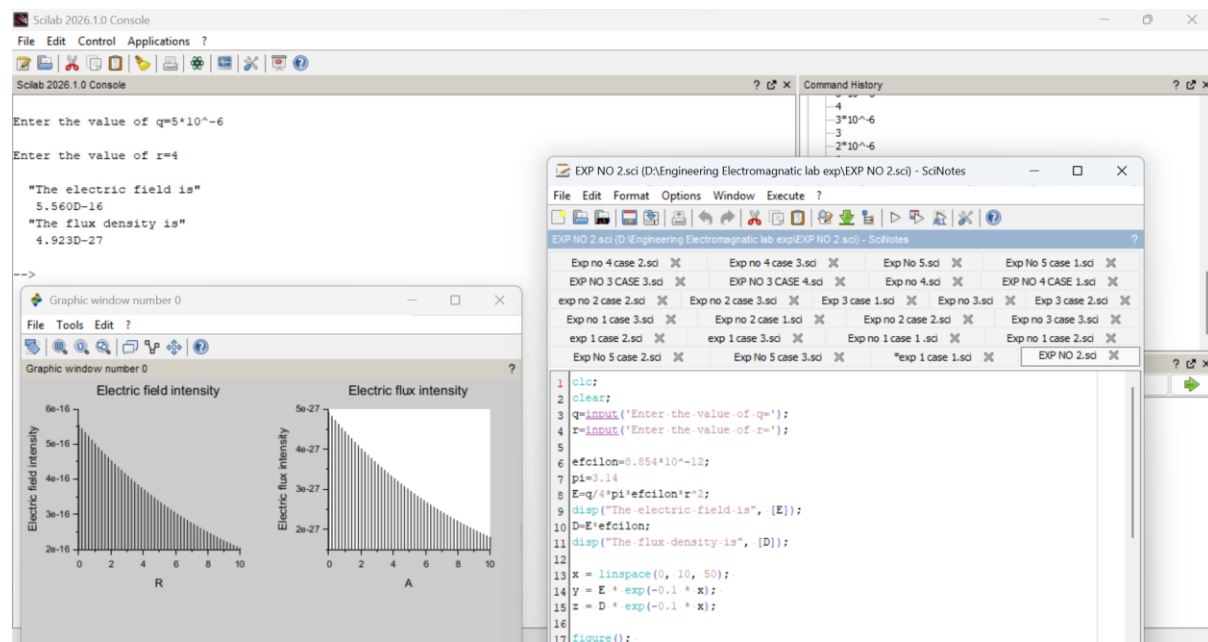
Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

Aim:

To write program for electric field and electric flux density in sphere using SCILAB.

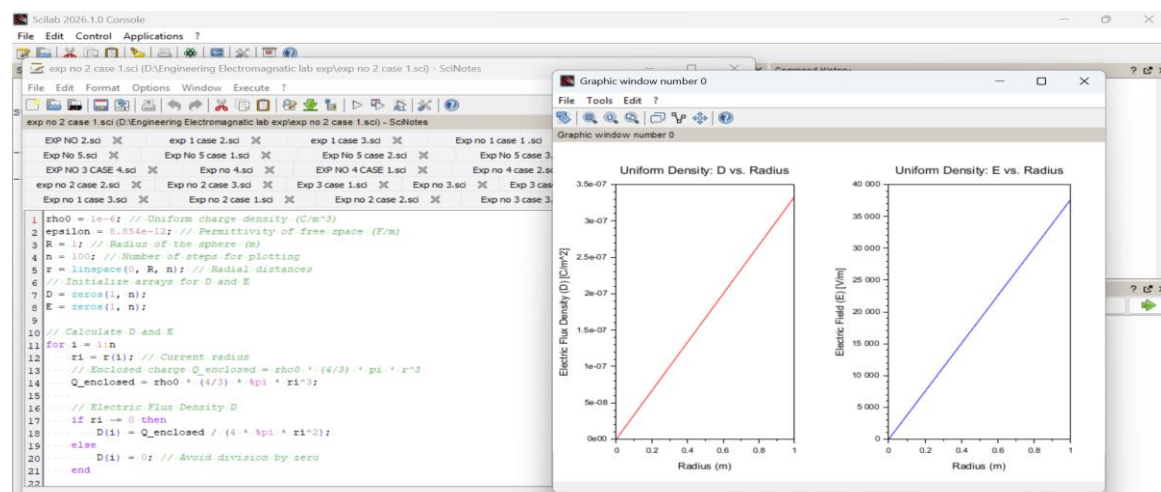
Software required:

- SCILAB version 6.0.



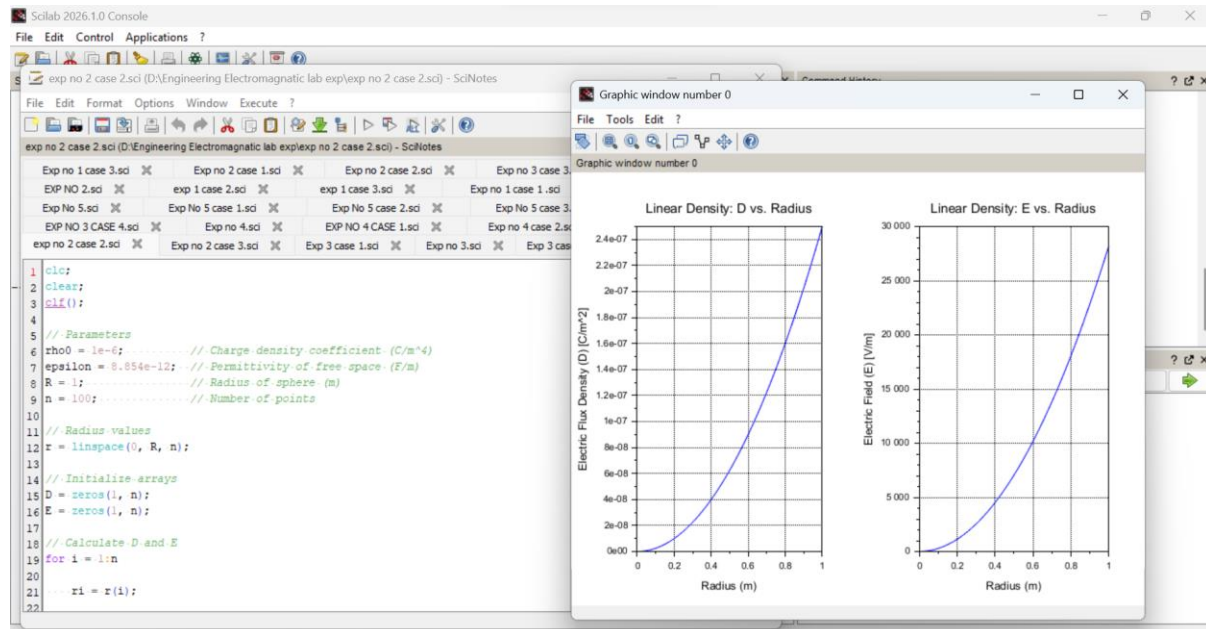
Case 1:

Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ_0 . What are the values of E and D at the center and at the surface of the sphere?



Case 2:

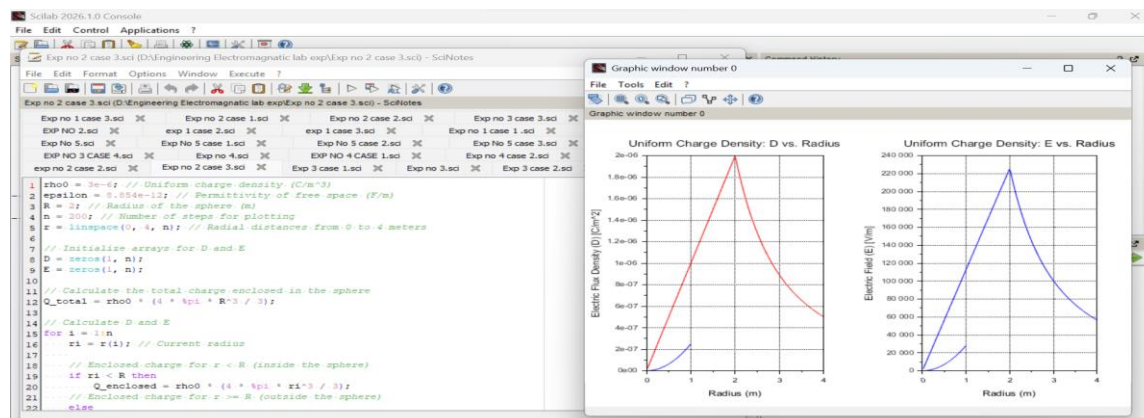
For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive the expressions for the electric field (E) and electric flux density (D). Plot their variations inside the sphere and analyze the trends.



Case 3:

A sphere of radius $R = 2$ m has a uniform charge density $\rho = 3 \times 10^{-6}$ C/m³

1. Calculate and plot the electric flux density $D(r)$ and electric field $E(r)$ as a function of radial distance r , ranging from 0 to 4 meters.
2. Discuss the behaviour of the electric field inside and outside the sphere.



Result:

Thus, the program was verified for electric field and electric flux density in sphere using SCILAB was successfully.

